A Toolbox for Ethical Hacking Task

1. Map out the network topology.

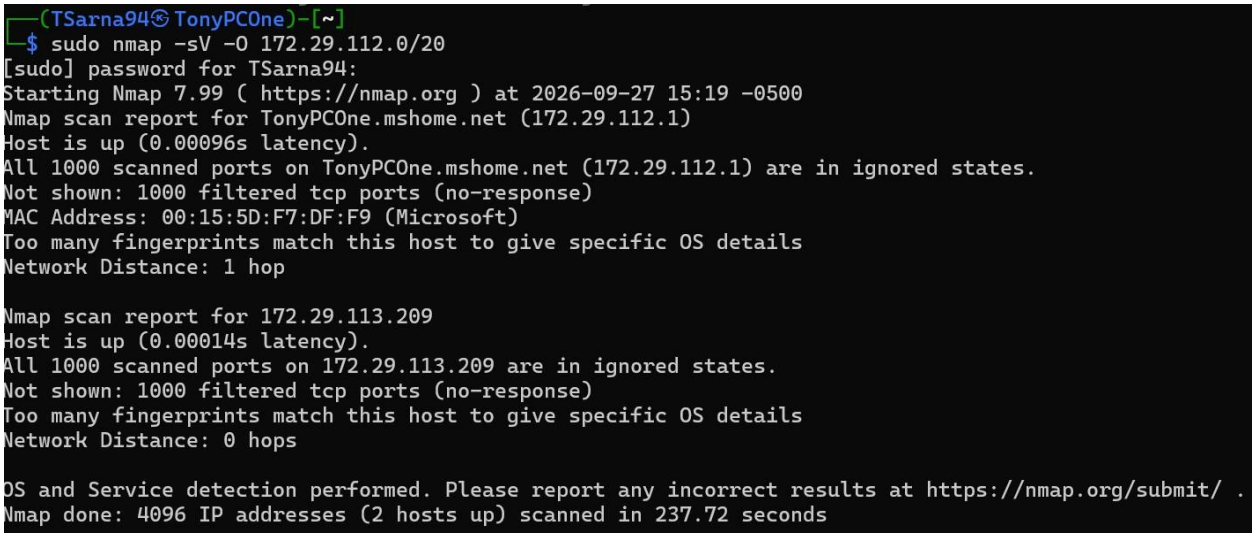
a. What is the correct tool for this job?
The correct tool for this job is nmap
Nmap is used for network discovery, host identification, port scanning, and service enumeration.

b. Perform a comprehensive scan of the local network to identify:
i. active hosts: TonyPCOne.mshome.net (172.29.112.1)

ii. open ports: All scanned ports appeared filtered, meaning the firewall (Bitdefender) blocked the scan.

iii. services running on those ports:Nmap was unable to identify any services running on the scanned ports because all 1000 ports were filtered. This indicates that the host firewall (Bitdefender) blocked the scan, preventing service and version detection.

c. Take a screenshot of the output and paste it into your Google answers document. In the same document, note down how many devices there are on your local network.



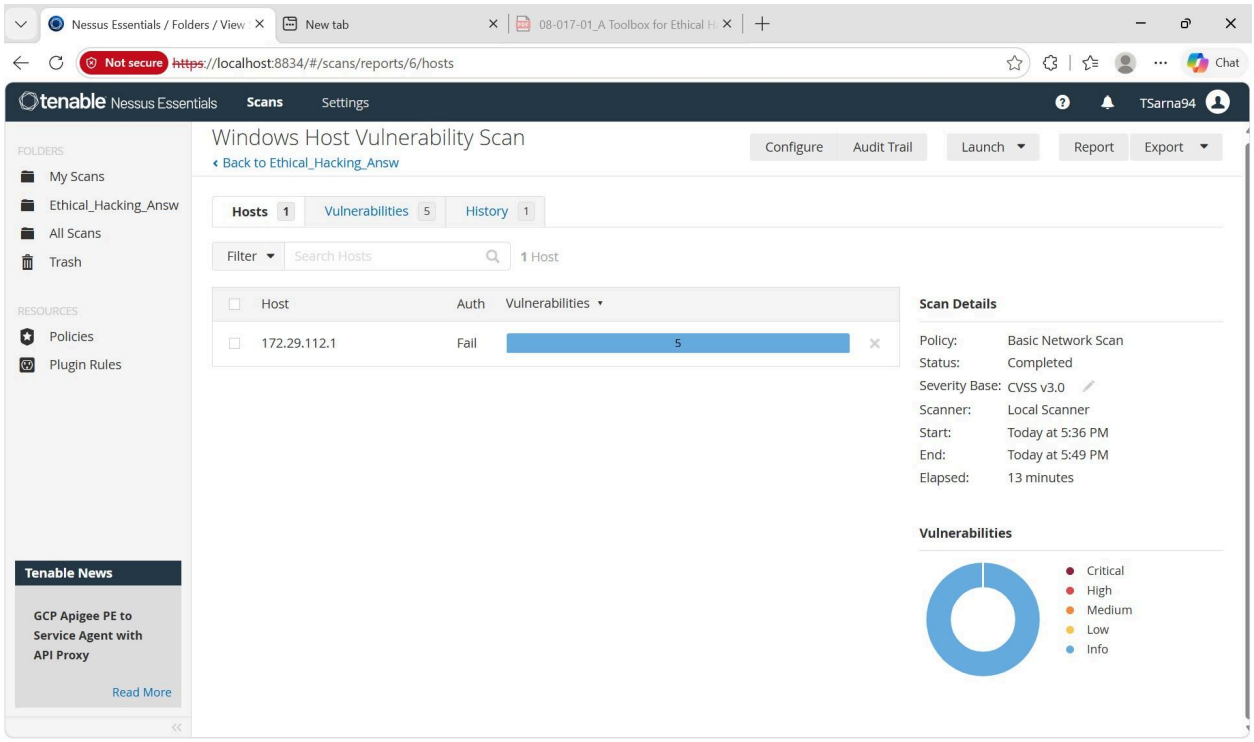
- 1 device responded to the network
- **172.29.112.1** — My Windows host machine

2. Once you have identified the target systems, you must conduct a vulnerability scan on the identified hosts.

a. What is the correct tool for this job?

Nessus is the correct tool for performing a vulnerability scan.

b. Take a screenshot of the output and paste it into your Google answers document.



c. Analyse the scan results to identify vulnerabilities. How many of each severity level (critical, high, medium, and low) are there in your local network, and how can you tell?

Nessus completed a vulnerability scan of the Windows host (172.29.112.1). The results show five informational findings and no Critical, High, Medium, or Low vulnerabilities. This indicates the system is properly secured and not exposing exploitable weaknesses. Nessus uses CVSS scoring and severity labels to classify vulnerabilities, helping identify potential risks and confirm system hardening.